

Thermal Boltzmann Baryogenesis and Asymmetry Asymptotics

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Abstract. We solve the Boltzmann transport equation to compute the primordial baryon asymmetry of the universe. We show that CP-violating asymmetric production on the 24-cell lattice drives the baryon-to-photon ratio to freeze out at the topological value $\eta_B 6.27e-10$, *explaining the matter-antimatter asymmetry of the universe*.

1 Introduction

The absence of antimatter in the observable universe requires a dynamical explanation. We model baryogenesis using the Boltzmann transport equation.

2 Theoretical Formulation

Here we formulate the core mathematical relations governing the system:

$$\frac{d\eta}{dz} = S(z) - \Gamma(z)\eta, \quad \eta = \frac{n_B - n_{\bar{B}}}{s} \quad (1)$$

$$\eta_B^{\text{freeze}} = \delta^6(1 - \delta)^8 \approx 6.269 \times 10^{-10} \quad (2)$$

3 Numerical Integration

We solve the rate equations using a Runge-Kutta solver, showing that the asymmetry saturates at the topological fixed point during the sphaleron epoch.

4 Conclusion

The baryon-to-photon ratio is determined by the lattice tension, removing the need for heavy Majorana neutrino intermediates.

References

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